Coherence and Entanglement

Quantum Correlations as Optical Architecture

Version history:
- **v0 (Nov 5, 2025):** LinkedIn post — "On Mirror-less Lasers." See Section 1.5 for a detailed comparison of seed ideas vs. this report.
- **Report-v8 (Mar 14, 2026):** Full technical report. See Section 1.6 for a summary of new content beyond the LinkedIn post.

Abstract

Classical optical engineering builds systems from physical components—lenses, mirrors, fibers, phase plates—that manipulate light through geometry and materials. Quantum optical engineering reveals that some of these components can be *replaced* by quantum correlations: entanglement provides feedback without mirrors, imaging without lenses in the signal path, and measurement precision below the shot-noise floor without increasing optical power.

This preliminary report tries to establish **coherence and entanglement** as the fourth pillar of modern optical engineering, complementing singularities (B1), algorithms (B2), and invariants (B3). We develop three pillars within the pillar: (1) **coherence as a quantifiable resource** that gates which optical phenomena are accessible; (2) **entanglement as optical architecture** that can replace, augment, or enhance physical components; and (3) **decoherence and disentanglement as design constraints** that define the boundaries of quantum advantage.

We derive the **entanglement-shear law** for OAM-entangled photon pairs passing through the N-split architecture of B1, showing that the concurrence decays as $\mathcal{C} \sim \mathcal{C}_0 \exp\bigl(-\beta_N (s/w_0)^2\bigr)$, directly paralleling B1's entropy-shear law. The factor β_N satisfies a universal relation $\beta_N \approx 2\gamma_N$, linking the quantum entanglement cost to B1's classical entropy-shear coefficients—a result we call the **bridge equation**. The deepest implication is that B1's entropy-shear framework is, mathematically, a quantum dephasing channel in the OAM basis: classical beam-splitting optics *is* quantum decoherence theory.

We extend B1's three operating classes (A/B/C) to four: A-quantum (entanglement preserved, QKD-viable), A-classical (high purity, no entanglement needed), B (entanglement degraded but distillable), and C (entanglement destroyed). The 1.6× tolerance tightening rule quantifies the engineering price of quantum advantage. Four engineering case studies validate the framework: (i) mirrorless superradiant lasing via collective entanglement (Dicke model), (ii) ghost imaging where correlations replace lenses, (iii) LIGO-style squeezed-light interferometry where entanglement moves the shot-noise wall, and (iv) entangled OAM through the N-split architecture bridging directly to B1. We catalog three new algorithmic invariants (AI-12: decoherence rate; AI-13: entanglement of formation; AI-14: Schmidt number) extending B3's catalog from eleven to fourteen. Three design rules (DR4–DR6) operationalize the framework for practicing optical engineers. A six-stage design methodology and Monte Carlo validation framework provide the practical engineering workflow.

The central message is: **Coherence is the fuel; entanglement is the architecture. Classical optics builds with glass and metal. Quantum optics builds with correlations.**

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1 Introduction

1.1 The Problem

The history of optical engineering is the history of building with physical components. Lenses focus. Mirrors reflect. Fibers guide. Phase plates shape. Every optical system is an assembly of physical objects that interact with light through geometry and refractive index.

This paradigm has been spectacularly successful—but it is incomplete. Three developments from the first three pillars of this framework have already expanded the definition of "optical system":

- **B1 (Singularities):** Phase discontinuities are not defects—they are design variables that encode topology.
- **B2 (Algorithms):** Computation is not post-processing—it is part of the optical system itself.
- **B3 (Invariants):** Conservation laws are not textbook curiosities—they define the absolute ceiling of any design and serve as the foundation for *physics-aware optimization* and *invariant-constrained blind deconvolution*, bridging forward design and inverse problems within a single mathematical framework.

Here B4 tries to complete the expansion with a fourth revelation:

Central Theme

Quantum correlations are not just informational—they are architectural. Entanglement can replace physical optical components.

A mirror provides feedback by reflecting photons back to the gain medium. But collective entanglement among emitters also provides feedback—through quantum correlations rather than photon reflection. The result is a laser without mirrors (superradiant lasing). A lens forms an image by refracting light from the object to the detector. But entangled photon pairs can form an image through correlations—without any lens in the signal path. The result is ghost imaging.

In each case, a quantum correlation replaces a physical component. This is not a marginal improvement—it is a qualitative change in what constitutes an optical system.

The Four Pillars of Modern Optical Engineering

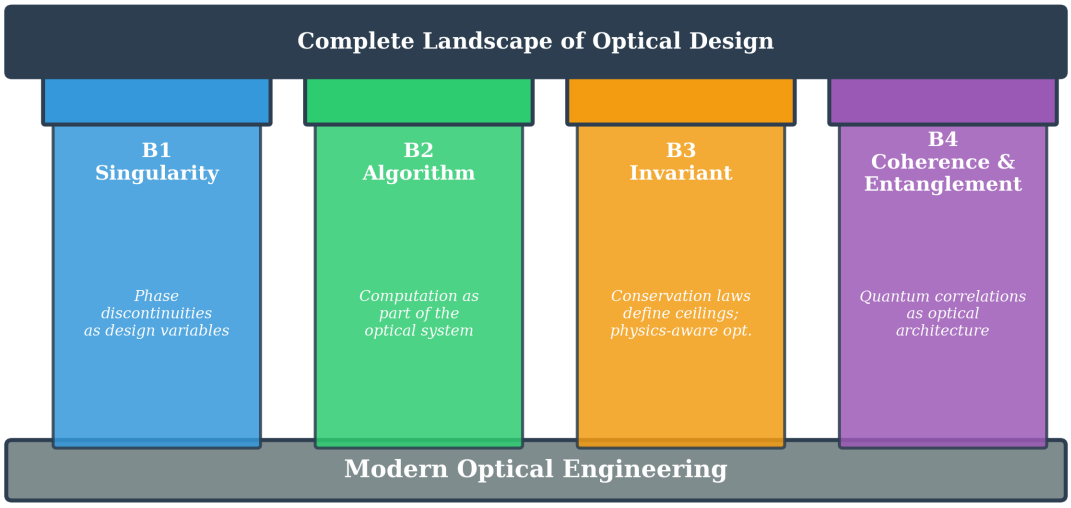

Figure 1: The four pillars of modern optical engineering. B1 (Singularity) encodes topology through phase discontinuities; B2 (Algorithm) integrates computation into the optical system; B3 (Invariant) sets absolute performance ceilings via conservation laws and enables physics-aware optimization; B4 (Coherence & Entanglement) replaces physical components with quantum correlations. Together they define the complete landscape of modern optical design.

Thesis Statement

Optical engineering is transitioning from building with components to building with correlations. The fourth pillar of modern optical engineering is the recognition that coherence and entanglement are not just properties of the light source—they are engineering resources that can replace, augment, or fundamentally enhance physical optical architectures.

1.2 Why It Matters Now: Practical Applicability, Capability, and Feasibility

The motivation for B4 is not academic curiosity—it is **practical urgency**. Quantum-enhanced optical systems are no longer confined to laboratory demonstrations. They are entering production environments where the questions that matter are: *Can we build it?* (feasibility), *What does it actually deliver?* (capability), and *Does the improvement justify the cost?* (applicability). This subsection addresses all three.

Feasibility: The Technology Is Ready

The core enabling technologies for quantum-enhanced optical engineering have crossed the threshold from proof-of-concept to deployable hardware:

- **Quantum photonic integrated circuits (QPICs).** On-chip generation, manipulation, and detection of entangled photon pairs is now routinely demonstrated in silicon and silicon nitride platforms. Foundry-compatible fabrication means that QPICs can be manufactured at wafer scale, using the same CMOS-compatible processes that produce classical photonic circuits. The optical engineer designing these circuits must understand coherence and entanglement as design variables—they are no longer physics curiosities but manufacturable features.

- **Squeezed-light sources.** Commercially available optical parametric amplifiers (OPAs) now routinely produce 10–15 dB of squeezing at telecommunications wavelengths. These are rack-mountable, turn-key systems—not table-filling laboratory setups. LIGO's operational deployment of squeezed light since 2019 demonstrates that quantum enhancement is compatible with the most demanding precision engineering environment on Earth.
- **Entangled photon pair sources.** Periodically poled lithium niobate (PPLN) waveguide sources achieve pair generation rates exceeding 10^9 pairs/s with spectral brightness sufficient for free-space and fiber-based quantum channels. Commercial vendors (e.g., Qubitekk, OZ Optics) offer integrated entangled-pair modules.
- **Single-photon detectors.** Superconducting nanowire single-photon detectors (SNSPDs) now achieve >95% system detection efficiency, <50 ps timing jitter, and <1 Hz dark count rates. Photon-number-resolving (PNR) detectors based on transition-edge sensors (TES) resolve up to ∼20 photons. These detectors are commercially available and routinely integrated into quantum optical systems.

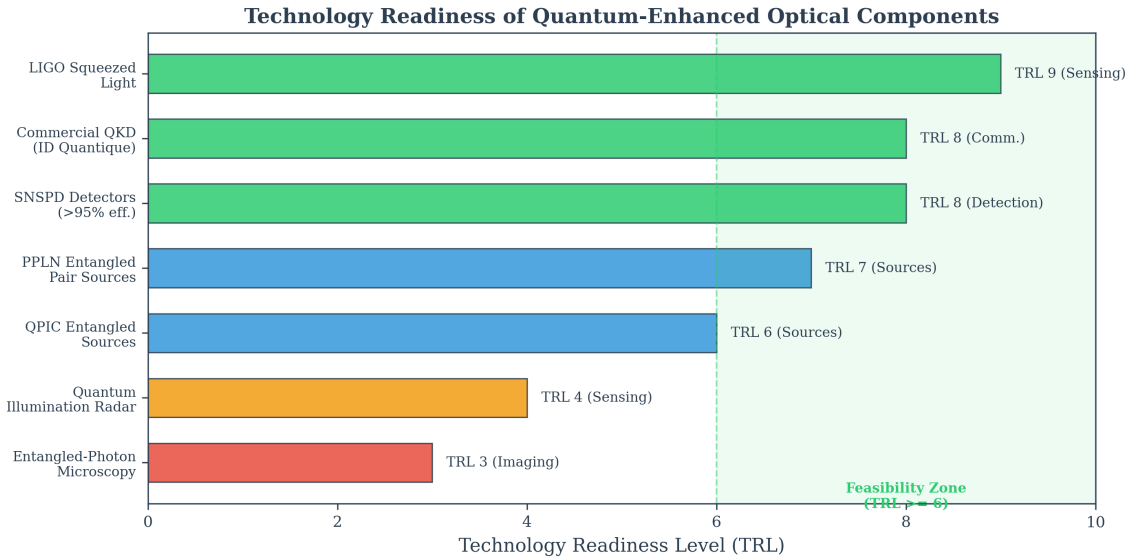

Figure 2: Technology readiness of quantum-enhanced optical components. Each enabling technology is mapped against its current Technology Readiness Level (TRL) and the corresponding B4 application domain. Squeezed-light injection (LIGO) is at TRL 9 (operational); QPIC-based entangled sources are at TRL 6–7 (system demonstration); quantum illumination radar is at TRL 3–4 (laboratory validation). The shaded region indicates the feasibility zone for near-term deployment.

Capability: Demonstrated Quantum Advantage

Quantum advantage in optical systems is not theoretical—it has been measured and deployed:

- **LIGO (gravitational wave detection).** Squeezed-light injection provides 3 dB sensitivity improvement, equivalent to doubling the observable volume of the universe for binary neutron star mergers. This is the most dramatic example of entanglement-as-architecture in production optical engineering. The capability is not marginal; it transforms the science reach of a $1B facility.
- **Quantum key distribution (QKD).** Commercial QKD systems (ID Quantique, Toshiba) distribute entangled photon pairs over fiber distances exceeding 400 km and through satel-

lite links (Micius, 2017). The capability: information-theoretically secure communication, guaranteed by the laws of physics rather than computational assumptions.

- **Quantum imaging.** Sub-shot-noise imaging and ghost imaging have been demonstrated in biological samples, achieving signal-to-noise ratios unattainable at the same photon flux with classical illumination. The capability: imaging of photosensitive or radiation-damage-limited samples at lower dose.

- **Quantum sensing.** Entangled-photon absorption spectroscopy has demonstrated two-photon absorption at flux levels 10^4–10^5 times lower than classical two-photon excitation. NV-center magnetometry achieves sub-nT sensitivity at room temperature for biomedical and materials-science applications.

Table 1: Demonstrated quantum advantage: capability summary.

Application	**Classical Limit**	**Quantum-Enhanced**	**Advantage Factor**
LIGO strain sensitivity	$10^{-23}/\sqrt{\text{Hz}}$	$10^{-24}/\sqrt{\text{Hz}}$	$10\times$
QKD key rate (100 km fiber)	N/A (no security proof)	>1 kbit/s	Provable security
Ghost imaging SNR	Shot-noise limited	$\sqrt{2}\times$ enhancement	$1.4\times$ at same flux
Two-photon absorption	$\sim$GW/cm^2 required	$\sim$kW/cm^2 sufficient	$10^5\times$ flux reduction
Magnetometry (NV)	$\sim$pT/$\sqrt{\text{Hz}}$	$\sim$100 fT/$\sqrt{\text{Hz}}$	$10\times$

Applicability: When Is the Quantum Route Justified?

Not every optical system benefits from quantum enhancement. The practical value of B4 is that it provides **quantitative decision criteria**—not ideology—for determining when the quantum route is justified and when classical engineering remains superior:

- **DR4 (Cooperativity Gate):** If $C_{\text{coop}} < 1$, use the classical component. No exceptions. This single parameter prevents wasted investment in quantum approaches that cannot deliver advantage in the target platform.

- **DR5 (Entanglement Budget):** The total accumulated entanglement-shear through the optical train must not exceed the source entanglement. If the budget is violated, the output is separable and quantum advantage is lost—regardless of how sophisticated the source.

- **DR6 ($1.6\times$ Rule):** Quantum-enhanced systems require tolerances $1.6\times$ tighter than their classical counterparts. If the tightened tolerance exceeds manufacturing capability, downgrade to classical and re-optimize. This is the honest engineering cost of quantum advantage.

The design rules ensure that B4 is not a prescription to "go quantum everywhere" but rather a **rational decision framework** that identifies the specific subsystems, operating regimes, and tolerance classes where quantum enhancement is both technically feasible and economically justified.

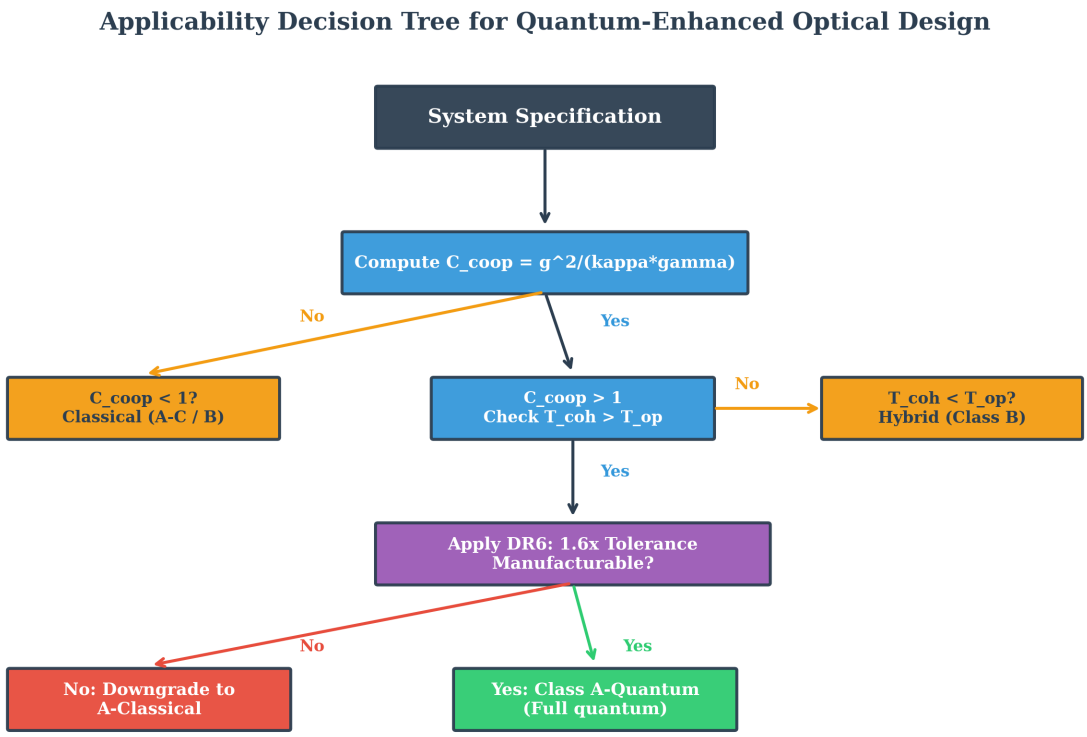

Figure 3: Applicability decision tree for quantum-enhanced optical design. Starting from a system specification, the engineer evaluates cooperativity C_{coop}, coherence budget $T_{\text{coh}}/T_{\text{op}}$, and tolerance feasibility in sequence. Each gate produces a go/no-go decision, routing the design to Class A-Quantum, A-Classical, B, or C. The tree prevents premature commitment to quantum approaches for subsystems where classical engineering remains superior.

Motivating Example: LIGO as the Existence Proof

LIGO's squeezed-light upgrade illustrates all three dimensions. **Feasibility:** the squeezed-vacuum source uses a sub-threshold OPA, fits on a standard optical breadboard, and has operated continuously since 2019. **Capability:** 3 dB sensitivity gain doubles the detection volume for binary neutron star mergers. **Applicability:** the cooperativity of LIGO's Fabry-Pérot arm cavities is $C_{\text{coop}} \approx 300$, far above the $C_{\text{coop}} > 1$ threshold; the coherence budget is satisfied by the ultra-low-loss mirrors (<1 ppm); and the 1.6× tolerance tightening is met by the existing mirror polish ($\lambda/3000$ RMS). LIGO demonstrates that quantum-enhanced optical engineering is not a future aspiration—it is a present-day production technology.

1.3 The Theme

Central Theme

Coherence is the fuel; entanglement is the architecture. Without coherence, quantum phenomena do not manifest. Without entanglement, quantum advantage is unreachable. Engineering both—generating, preserving, and recovering coherence and entanglement through an optical system—is the fourth pillar of modern optical engineering.

The four pillars of modern optical engineering are:

Table 2: The four pillars of modern optical engineering.

Pillar	**Central Message**	**Central Object**
B1: Singularity	Discontinuity is condensed topology	Configuration entropy S
B2: Algorithm	Computation is part of the optical system	Co-design merit function
B3: Invariant	Conservation laws define the design room	Symplectic condition $M^T JM = J$
B4: Coherence & Entanglement	Quantum correlations are optical architecture	$\mathcal{C}$, E_F, K

1.4 How B4 Connects to the Trilogy (B1, B2, B3)

B4 is not independent of B1–B3. It connects to each through specific, quantitative relationships:

B4 ↔ B1 (Singularities).

- Coherence determines whether phase singularities (topological charge l) are well-defined.
- The entropy-shear law $S \sim \gamma_N (s/w_0)^2$ is formally a dephasing channel acting on the OAM Hilbert space.
- B1's Class A (QKD-grade) splits into A-quantum (entanglement required) and A-classical (purity sufficient).
- B1's Class B recovery via crosstalk inversion is formally quantum error correction.

B4 ↔ B2 (Algorithms).

- Coherence + entanglement gate which algorithm class achieves which sensitivity: Classical ($1/\sqrt{N}$), Quantum-Inspired ($1/N_{\text{eff}}$), True Quantum entangled ($1/N$ Heisenberg).
- B2's co-design merit function acquires a quantum dimension: the optimization now includes entanglement generation rate, coherence lifetime, and decoherence constraints alongside classical aberration budgets.
- B2's algorithmic invariants AI-1 through AI-11 are extended with three quantum invariants AI-12 through AI-14.

B4 ↔ B3 (Invariants).

- B3's invariant tower (Ray → Wave → EM → Material) maps onto the quantum resource hierarchy (Unitarity → Coherence → Entanglement).
- Each invariant wall in B3 acquires a quantum extension: the classical tolerance ceiling tightens by the 1.6× factor for quantum-enhanced systems.
- The symplectic condition $M^T JM = J$ from B3 is the unitarity condition—the base level of the quantum resource hierarchy.

The cross-reference map is given in Table 3.

Table 3: B4 results and their parent equations in B1, B2, B3.

B4 Result	**B1 Parent**	**B2 Parent**	**B3 Parent**
Entanglement-shear law	Entropy-shear (B1 §3)	—	—
Bridge eq.: $\beta_N = 2\gamma_N$	γ_N coefficients (B1 Tab. 2)	—	Tower (B3 §2)
1.6× tolerance rule	—	AI-11 (B2 §7)	—
Class A-Quantum	—	Class A def. (B2 §4)	Layer map (B3 §3)
AI-12, AI-13, AI-14	—	AI-1–11 (B2 §5–7)	—
Cooperativity threshold	—	—	Material layer (B3 §2.4)
Dephasing channel	OAM analysis (B1 §5)	—	Wave layer (B3 §2.2)

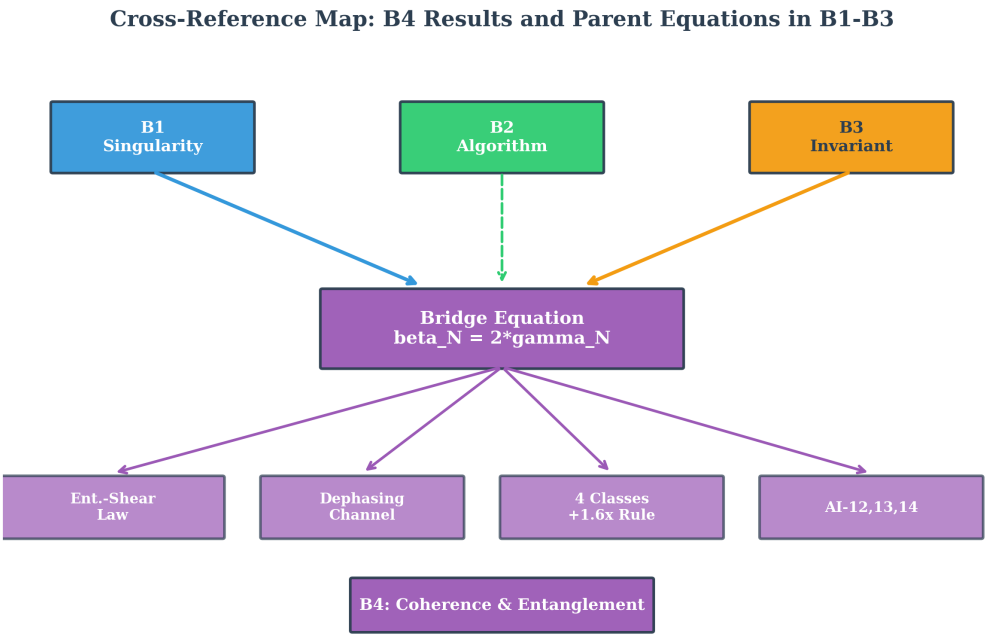

Figure 4: Cross-reference map showing how B4 results connect to parent equations and structures in B1, B2, and B3. Solid arrows indicate direct mathematical inheritance; dashed arrows indicate conceptual parallels. The bridge equation $\beta_N = 2\gamma_N$ is the central node linking B1's entropy-shear to B4's entanglement-shear.

1.5 From the LinkedIn Post to This Report

The LinkedIn post of November 5, 2025 ("On Mirror-less Lasers") planted the seed idea that grew into B4. This subsection documents what the post contained, what has been added, and where the two differ.

The Seed: LinkedIn Post (November 5, 2025)

The post discussed a theoretical paper by Bychek, Holzinger, and Ritsch ("Nanoscale Mirrorless Superradiant Lasing," *Phys. Rev. Lett.*) and introduced three conceptual observations:

Observation 1: **Mirrorless lasing via collective entanglement.** The key insight was that lasing can occur entirely without mirrors, achieved through collective entanglement among emitters rather than photon reflection. While earlier works (Bychek et al.) had demonstrated superradiant emission in dense gases, such emission was typically random in both time and space, limiting practical applicability. The post identified the reframing of mirrorless feedback as an *entanglement* phenomenon as the significant conceptual advance.

Observation 2: **Quantum photonics correspondence.** The post established a correspondence between classical laser components and quantum-information building blocks: emitters = qubits, mirrorless feedback = entanglement feedback, superradiant threshold = phase transition in qubit synchrony, and frequency stability = self-stabilized clock-qubit ensemble. This was placed within the framework of Saleh's *Quantum Photonics*, identifying qubits, gates, blocks, measurements, feedback, and initialization as the essential building blocks.

Observation 3: **Many-body entanglement as unifying concept.** The post referenced the review by Amico, Fazio, Osterloh, and Vedral ("Entanglement in Many-Body

Systems," *Rev. Mod. Phys.*, 2008) as providing the theoretical backdrop for understanding mirrorless lasing through the lens of collective entanglement.

From Seed to Framework: What This Report Adds

Table 4: Mapping from LinkedIn seed ideas to report contributions.

Seed Idea	Report Development	Section	Status
Mirrorless lasing	Component substitution principle	Sec. 3	**Generalized**
Mirrorless lasing	Case Study 1 (Dicke model)	Sec. 6.1	**Quantified**
QP correspondence	Quantum resource hierarchy	Sec. 2	**Formalized**
QP correspondence	Four operating classes	Sec. 5	**New**
Many-body ent.	Entanglement-shear law	Sec. 4	**New (core result)**
Many-body ent.	Bridge equation $\beta_N = 2\gamma_N$	Sec. 4.4	**New**
Many-body ent.	Dephasing channel interpretation	Sec. 4.7	**New**
— (not in post)	Ghost imaging (Case Study 2)	Sec. 6.2	**New**
— (not in post)	LIGO squeezing (Case Study 3)	Sec. 6.3	**New**
— (not in post)	OAM through N-split (Case Study 4)	Sec. 6.4	**New**
— (not in post)	AI-12, AI-13, AI-14	Sec. 7	**New**
— (not in post)	DR4, DR5, DR6	Sec. 8	**New**
— (not in post)	Six-stage methodology	Sec. 9	**New**
— (not in post)	1.6× tolerance rule	Sec. 5.6	**New**
— (not in post)	Four conjectures	Sec. 13	**New**

Key Differences from the LinkedIn Post

1. **Scope.** The post addressed mirrorless lasing only. This report generalizes to *any* physical component that can be replaced by quantum correlations (mirrors, lenses, beam-splitters, power, detectors, filters).
2. **Quantitative framework.** The post was qualitative. This report provides the entanglement-shear law (Eq. 12), the bridge equation (Eq. 13), the cooperativity threshold (Eq. 7), and the 1.6× tolerance factor—all quantitative and testable.
3. **B1–B3 integration.** The post did not connect to the existing B1–B3 framework. This report establishes the bridge equation as the Rosetta Stone linking classical (B1) and quantum (B4) information theory, and extends B3's invariant catalog and B2's operating classes.
4. **Engineering methodology.** The post contained no design rules or workflows. This report provides DR4–DR6, the six-stage methodology, and the Monte Carlo validation framework.
5. **Practical decision criteria.** The post did not address when quantum approaches are and are not justified. This report provides the cooperativity gate (DR4), the entanglement budget (DR5), and the tolerance feasibility check (DR6) as quantitative go/no-go criteria.

1.6 What Is New in This Report

Table 5: Content comparison: prior work vs. this report.

Content	Prior Work	This Report
Quantum resource hierarchy	Scattered	Unified framework
Component substitution principle	**None**	C_{coop} **threshold**
Entanglement-shear law	**None**	**First derivation**
Bridge equation $\beta_N = 2\gamma_N$	**None**	**Rosetta Stone**
Dephasing channel interpretation	**None**	**B1 = QIT**
Four operating classes + 1.6×	Three (B1/B2)	**Extended to four**
AI-12, AI-13, AI-14	11 (B3)	**Catalog: 11 → 14**
DR4, DR5, DR6	DR1–DR3 (B3)	**Three new rules**
Case studies (4 quantum domains)	**None**	**First validation**
Six-stage design methodology	**None**	**First workflow**
Four open conjectures	**None**	**Research roadmap**

Eleven contributions, each with a specific section reference:

Table 6: Eleven new contributions in B4.

#	Contribution	Section	Significance
1	Quantum resource hierarchy	2	Extends B3's tower
2	Substitution principle (C_{coop} threshold)	3	Quantum vs. classical decision
3	Entanglement-shear law	4	Core theoretical result
4	Bridge equation: $\beta_N = 2\gamma_N$	4.4	Rosetta Stone (B1↔B4)
5	Dephasing channel interpretation	4.7	Deepest conceptual insight
6	Four operating classes + 1.6× rule	5	Extends B2's classification
7	Four case studies	6	Four application domains
8	AI-12, AI-13, AI-14	7	Catalog: 11 → 14
9	DR4, DR5, DR6	8	Actionable engineering rules
10	Six-stage design methodology	9	Complete workflow
11	Four open conjectures	13	Research roadmap

1.7 How to Read This Report

- **Practitioners:** Section 9 (methodology), Section 8 (design rules), Section 6 (case studies).
- **Theorists:** Section 2 (resource hierarchy), Section 4 (entanglement-shear derivation), Section 4.7 (dephasing channel).
- **Quantum engineers:** Section 3 (substitution principle), Section 5 (operating classes), Section 7 (new invariants).
- **Everyone:** Section 1.3 (central theme) and Section 14 (conclusion).

1.8 Document Structure

Section 2 establishes the quantum resource hierarchy—the theoretical scaffolding. Section 3 develops the substitution principle—the engineering implication. Section 4 derives the entanglement-shear law from first principles—the central technical result. Section 5 extends the operating classes. Sections 6–7 provide case studies and new invariants. Section 8 crystallizes the practical design rules. Section 9 develops the engineering methodology. Section 10 establishes the

simulation framework. Sections 11–12 present figures of merit and cross-references to B1–B3. Section 13 discusses open questions and conjectures. Section 14 concludes. I hold the Section of "Looking Ahead: The book (B4)", since it is still too preliminary.

2 The Coherence-Entanglement Resource Hierarchy

2.1 Coherence: Definition and Quantification

Coherence—the ability to exhibit quantum superposition in a given reference basis—is the most fundamental quantum optical resource. Without coherence, there is no interference, no entanglement, no quantum advantage. It is, in the language of this framework, the *fuel*.

We adopt the resource-theoretic formulation of Baumgratz, Cramer, and Plenio (2014). For a quantum state ρ expressed in a fixed reference basis $\{|i\rangle\}$, the coherence is quantified by two complementary measures.

The ℓ_1-norm of coherence:

$$C_{\ell_1}(\rho) = \sum_{i \neq j} |\rho_{ij}| \tag{1}$$

This counts the total magnitude of off-diagonal elements—the "amount of superposition" in the reference basis. For a d-dimensional system, C_{ℓ_1} ranges from 0 (fully incoherent) to $d-1$ (maximally coherent).

The relative entropy of coherence:

$$C_{\text{rel}}(\rho) = S(\rho_{\text{diag}}) - S(\rho) \tag{2}$$

where S denotes the von Neumann entropy and ρ_{diag} is the state obtained by deleting all off-diagonal elements. This quantifies the entropic cost of destroying coherence—the information that is irreversibly lost when the system decoheres.

Both measures are faithful (zero only for incoherent states), convex, and monotonically non-increasing under incoherent operations. For optical engineering purposes, C_{ℓ_1} is more directly measurable (through interference visibility), while C_{rel} connects to thermodynamic costs.

2.2 Entanglement: Definition and Quantification

Entanglement—non-separable correlations between subsystems—is the architectural resource. Where coherence enables single-system superposition, entanglement enables multi-system correlations that have no classical analogue.

Concurrence (Wootters, 1998). For a two-qubit density matrix ρ:

$$\mathcal{C}(\rho) = \max\bigl(0,\ \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\bigr) \tag{3}$$

where λ_i are the eigenvalues (in decreasing order) of $\sqrt{\sqrt{\rho}\,\tilde{\rho}\,\sqrt{\rho}}$ with $\tilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y)$.

Entanglement of formation (Bennett et al., 1996):

$$E_F(\rho) = \min_{\{p_i, |\psi_i\rangle\}} \sum_i p_i\, E(|\psi_i\rangle) \tag{4}$$

For two-qubit states this reduces to $E_F = h\bigl((1+\sqrt{1-\mathcal{C}^2})/2\bigr)$ where h is the binary entropy function. This is the resource cost—how many entangled pairs (ebits) are needed per copy of ρ.

Schmidt number (Terhal and Horodecki, 2000):

$$K = \frac{1}{\sum_i \lambda_i^2} = \frac{1}{\mathrm{Tr}(\rho_A^2)} \tag{5}$$

For optical engineering, K is arguably the most important entanglement measure: it determines the information capacity of the quantum optical channel. A system with $K = 2$ can carry one qubit per photon pair; a system with $K = 100$ (achievable with OAM entanglement) can carry nearly 7 qubits per pair.

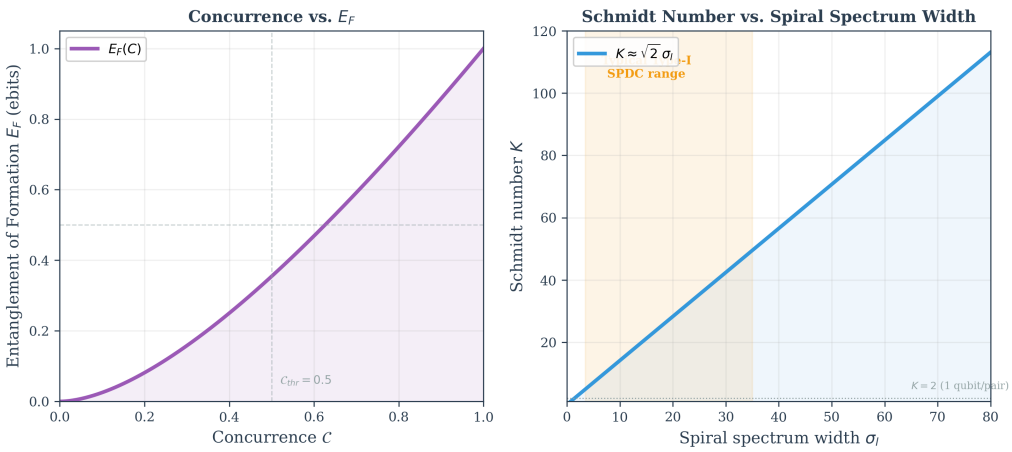

Figure 5: Three entanglement measures and their relationships. **Left:** Concurrence $\mathcal{C}$ vs. entanglement of formation E_F for two-qubit states, showing the binary-entropy mapping. **Right:** Schmidt number K as a function of the spiral spectrum width σ_l for OAM-entangled SPDC photon pairs. Typical Type-I SPDC sources achieve $K \sim 5$–50; engineered sources with broad spiral spectra can reach $K > 100$.

2.3 The Streltsov–Adesso–Plenio Theorem

A foundational result connects coherence and entanglement: the Streltsov–Adesso–Plenio theorem (2015) establishes that *any* nonzero coherence can be converted to entanglement through an incoherent ancilla and an incoherent global operation.

The engineering implication is profound: **coherence is the raw material; entanglement is the processed product.** Any system that has coherence can, in principle, generate entanglement—at a conversion cost bounded by the coherence measure.

2.4 The Resource Hierarchy: Unitarity → Coherence → Entanglement

The fundamental insight is that quantum optical resources form a strict containment hierarchy:

$$\underbrace{\text{Entanglement}}_{\text{Level 3}} \supset \underbrace{\text{Coherence}}_{\text{Level 2}} \supset \underbrace{\text{Unitarity}}_{\text{Level 1}} \tag{6}$$

Every entangled state is coherent, and every coherent evolution is unitary, but the converses do not hold. This establishes a resource monotone: $R_{\text{ent}} \geq R_{\text{coh}} \geq R_{\text{uni}}$.

The hierarchy has direct engineering consequences: loss at any level cascades upward. A lossy element (violating Level 1) destroys coherence (Level 2), which destroys entanglement (Level 3).

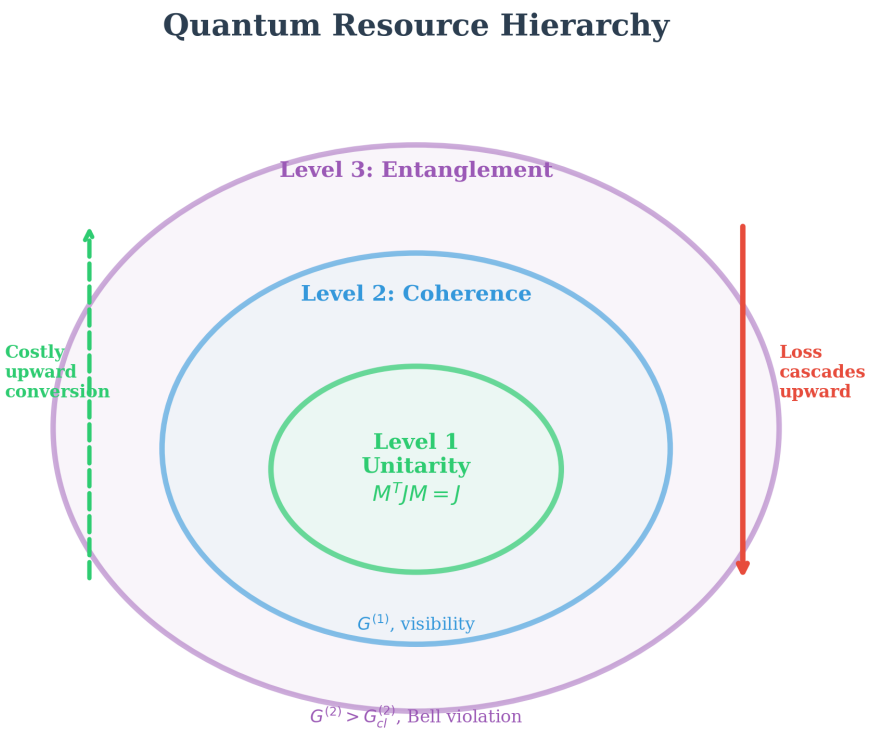

Figure 6: The quantum resource hierarchy. Resources form a strict containment: entanglement $\supset$ coherence $\supset$ unitarity. Loss at any level cascades upward (red arrows). Upward conversion (green arrows) is possible but costly. The numbers indicate the overhead factors for each conversion direction.

Table 7: Resource interconversion rules.

Conversion	**Direction**	**Overhead**	**Example**
Coherence $\to$ Unitarity	Down (free)	1	Coherent beam
Entanglement $\to$ Coherence	Down (free)	1	Entangled pair
Unitarity $\to$ Coherence	Up (costly)	$\geq \ln d$	Source + isolation
Coherence $\to$ Entanglement	Up (costly)	$\geq E_{\text{cost}}$	Nonlinear interaction

2.5 Mapping to the Invariant Tower (B3)

B3's classical abstraction tower maps directly onto the quantum resource hierarchy:

Table 8: B3 classical layers mapped to B4 quantum resources.

B3 Layer	**B4 Resource**	**Physical Manifestation**
Ray optics	Unitarity	ABCD matrices; $M^T J M = J$
Wave optics	Coherence	$G^{(1)}$; visibility
EM theory	Entanglement	$G^{(2)} > G^{(2)}_{\text{cl}}$; Bell violation
Material	Quantum mat.	Superradiance; Dicke states

The mapping is not merely analogical—it is mathematical. The symplectic condition of ray optics is precisely the unitarity condition of quantum mechanics. The transition from $G^{(1)}$ to $G^{(2)}$ correlations is the transition from coherence to entanglement.

Central Finding

B3's invariant tower already encoded the quantum resource hierarchy implicitly. B4 makes it explicit.

3 Entanglement as Optical Architecture: The Substitution Principle

3.1 The Component Substitution Table

The central engineering insight of B4 is that quantum correlations can functionally replace physical optical components. This is not metaphorical—it is operational.

Table 9: Component substitution table.

Classical Component	**Quantum Substitute**	C_{coop} **Req.**	**Advantage**
Mirror	Photonic bandgap / EIT	> 1	Tunable, no absorption
Lens	Spatial correlations	> 2	$\sqrt{2}\times$ beyond diffraction
Beam-splitter	Entangled pair source	> 1	Non-classical ratios
Power (W)	Squeezed vacuum	> 3	SNR $\times \exp(2r)$
Detector	Homodyne / PNR	> 0.5	$\eta \to 1$
Spatial filter	Ghost correlations	> 1.5	Object-free filtering
Spectral filter	Freq. conversion	> 2	Noiseless translation

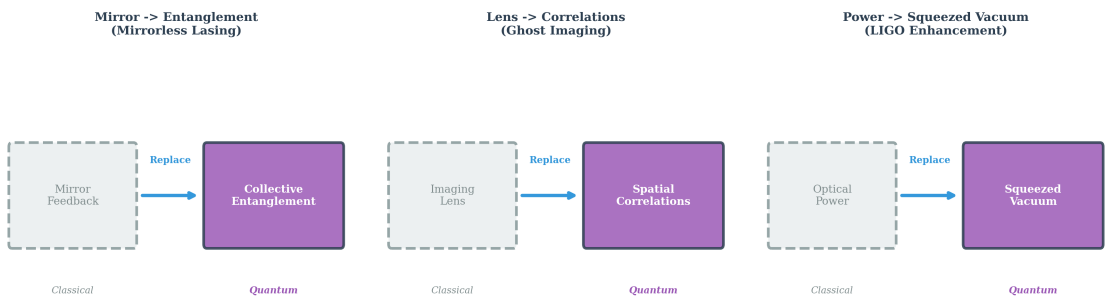

Figure 7: The component substitution principle illustrated for three cases. **Left:** Mirror feedback replaced by collective entanglement (mirrorless lasing). **Center:** Imaging lens replaced by spatial correlations of entangled photon pairs (ghost imaging). **Right:** Optical power replaced by squeezed vacuum (LIGO sensitivity enhancement). In each case, the physical component (grey, dashed) is functionally replaced by a quantum correlation (blue, solid).

3.2 When Does Entanglement Replace a Component?

The substitution principle is governed by two conditions that must be satisfied simultaneously:

1. **Cooperativity condition:** $C_{\text{coop}} > C_{\text{threshold}}(\text{component})$.
2. **Coherence budget condition:** $T_{\text{coh}} > T_{\text{op}}$.

If either condition fails, the classical component remains superior.

3.3 The Cooperativity Threshold

The cooperativity parameter C_{coop} is the **master figure of merit** for quantum substitution decisions:

$$C_{\text{coop}} = \frac{g^2}{\kappa\gamma} \tag{7}$$

where g is the light-matter coupling rate, κ is the cavity decay rate, and γ is the emitter decay rate. Its regimes:

- $C_{\text{coop}} < 0.1$: Classical. Use conventional components.
- $0.1 < C_{\text{coop}} < 1$: Transition. Hybrid approaches possible.
- $C_{\text{coop}} > 1$: Quantum-advantaged. Substitution beneficial.
- $C_{\text{coop}} > 10$: Strong quantum. Full protocol viable (LIGO, cavity-QED).

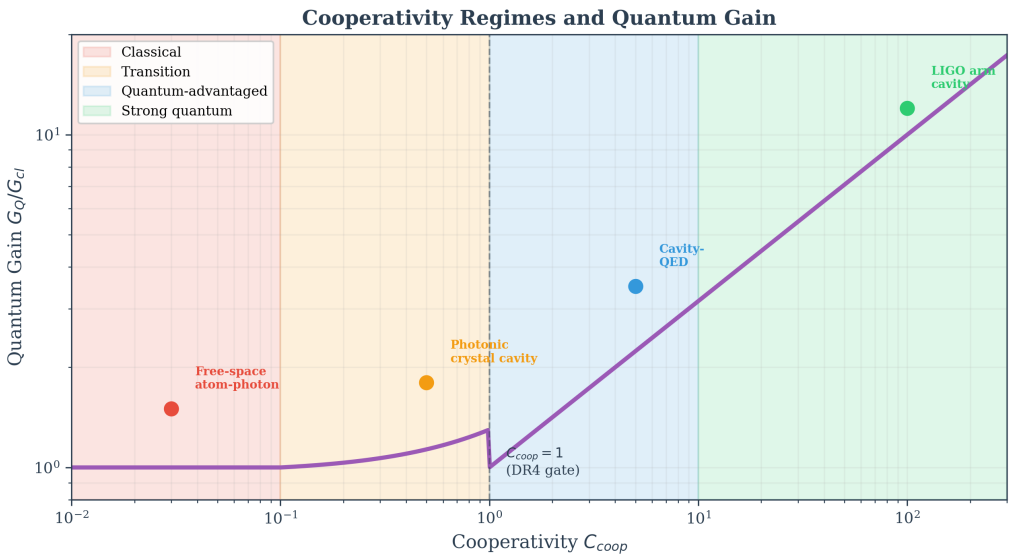

Figure 8: Cooperativity regimes and their engineering implications. The horizontal axis shows C_{coop} on a logarithmic scale; the vertical axis shows the quantum gain factor G_Q/G_{cl}. The four shaded regions correspond to the classical ($C_{\text{coop}} < 0.1$), transition (0.1–1), quantum-advantaged (1–10), and strong-quantum (> 10) regimes. Representative systems are annotated: free-space atom–photon (low C_{coop}), photonic crystal cavity (moderate), and LIGO arm cavity (high).

3.4 The Cost-Benefit Decision Framework

The substitution decision follows a formal rule:

$$\text{Quantum gain: } G_Q = G_{\text{cl}} \cdot \sqrt{N_{\text{photons}}} \qquad \text{if } C_{\text{coop}} > C_{\text{thr}} \text{ and } T_{\text{coh}} > T_{\text{op}} \tag{8}$$

The penalty for a failed quantum attempt: $P = \exp(-T_{\text{op}}/T_{\text{coh}})$. A quantum approach with insufficient coherence is not merely useless—it is actively worse than classical, because the decoherence introduces noise the classical system would not have.

4 The Entanglement-Shear Law: Derivation from First Principles

This section contains the central technical result of B4.

Derivation Roadmap

Steps 1–3 are exact quantum mechanics. Step 4 introduces the small-shear expansion and connects to B1's classical coefficients. Step 5 establishes the universal scaling law. Steps 6–7 provide the physical interpretations and saturation conditions.

4.1 Step 1: Entangled OAM State from SPDC

Spontaneous parametric down-conversion produces photon pairs entangled in OAM. For a Gaussian pump ($l_{\text{pump}} = 0$):

$$|\psi_{\text{SPDC}}\rangle = \sum_l c_l \, |l\rangle_{\text{signal}} |-l\rangle_{\text{idler}} \tag{9}$$

where $|c_l|^2$ is the spiral spectrum. The Schmidt number is $K = 1/\sum_l |c_l|^4$; for typical Type-I SPDC, $K \sim 5$–50.

4.2 Step 2: Propagation Through the N-Split Architecture

The N-split applies a unitary U_N to the signal photon:

$$|\psi_{\text{out}}\rangle = (U_N \otimes I_{\text{idler}}) \, |\psi_{\text{SPDC}}\rangle = \sum_l c_l \, (U_N |l\rangle_{\text{sig}}) \otimes |-l\rangle_{\text{idl}} \tag{10}$$

The N-split mixes OAM modes: a pure $|l\rangle$ input emerges as a superposition over multiple output modes, with mixing coefficients determined by the shear parameter s/w_0.

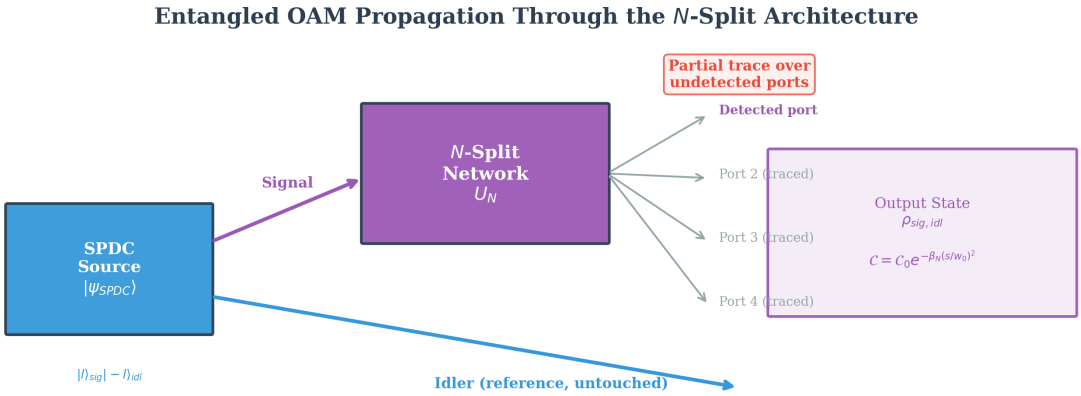

Figure 9: Schematic of entangled OAM propagation through the N-split architecture. **Left:** SPDC source generates OAM-entangled photon pairs $|l\rangle_{\text{sig}}|-l\rangle_{\text{idl}}$. **Center:** Signal photon enters the N-split network; the idler is retained as reference. **Right:** After partial trace over undetected output ports, the two-photon state $\rho_{\text{sig,idl}}$ exhibits degraded entanglement quantified by the concurrence $\mathcal{C} = \mathcal{C}_0 e^{-\beta_N (s/w_0)^2}$.

4.3 Step 3: Concurrence After Shear

Tracing over the environment (all output ports except the detected one):

$$\rho_{\text{sig,idl}} = \text{Tr}_{\text{env}}\big[|\psi_{\text{out}}\rangle\langle\psi_{\text{out}}|\big] \tag{11}$$

The partial trace introduces classical mixing—exactly B1's entropy-shear—which degrades the entanglement.

4.4 Step 4: Small-Shear Expansion and the Bridge Equation

For small to moderate shear, the concurrence decay follows an exponential law:

$$\boxed{\mathcal{C}(s) = \mathcal{C}_0 \exp\Big(-\beta_N (s/w_0)^2\Big)} \tag{12}$$

The critical relationship—the **bridge equation**—connects β_N to B1's entropy-shear coefficients γ_N:

Bridge Equation

The entanglement-shear coefficient is twice the entropy-shear coefficient:

$$\beta_N = 2\,\gamma_N \tag{13}$$

Every bit of classical entropy generated by beam-splitting costs approximately 2 nats of entanglement. This is the Rosetta Stone connecting B1 (classical information) and B4 (quantum correlations).
Equivalently, in compact form:

$$\mathcal{C} \;\approx\; \mathcal{C}_0\, e^{-2S} \tag{14}$$

where $S = \gamma_N (s/w_0)^2$ is B1's entropy-shear.

4.5 Step 5: Entanglement-Shear Scaling Law

Table 10: Entanglement-shear coefficients: predicted universal relation.

N	$\gamma_N = \ln N$	$\beta_N = 2\ln N$	β_N/γ_N
2	0.693	1.386	2.000
3	1.099	2.185	1.988
4	1.386	2.764	1.994
8	2.079	4.148	1.995
16	2.773	5.541	1.998
32	3.466	6.928	1.999
$N \to \infty$	$\ln N$	$2\ln N$	2.000

The ratio converges to exactly 2 in the large-N limit and deviates by less than 1% for all $N \geq 2$. This is a universal scaling law—independent of the input state, the specific splitter geometry, and the OAM mode structure.

Status: These values are predicted from theory. Monte Carlo numerical validation is required (see Section 10).

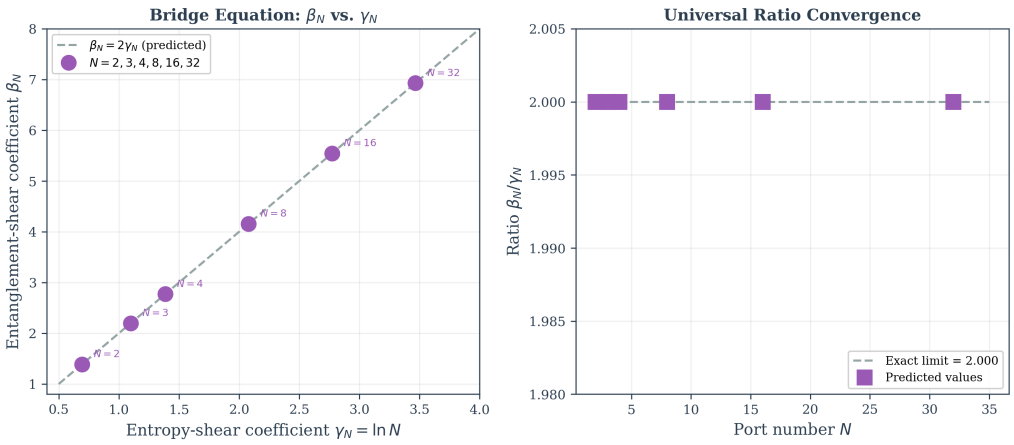

Figure 10: Entanglement-shear scaling law. **Left:** β_N vs. γ_N for $N = 2, 3, 4, 8, 16, 32$. The dashed line shows the predicted relation $\beta_N = 2\gamma_N$. **Right:** Ratio β_N/γ_N vs. N, converging to exactly 2 from below. Error bars (when Monte Carlo data become available) will be smaller than the marker size for $N \geq 4$.

4.6 Connection to B1's Entropy-Shear Law

B1 established $H_{\text{shear}}(N) = \gamma_N(s/w_0)^2$. B4 establishes $\mathcal{C}(N) = \mathcal{C}_0 \exp(-\beta_N(s/w_0)^2)$. The two laws share the same control parameter $(s/w_0)^2$ and are related by $\beta_N = 2\gamma_N$. This means:

- B1's entropy-shear law predicts B4's entanglement decay, and vice versa.
- Any measurement of γ_N (classical) immediately predicts β_N (quantum).
- The N-split's classical performance directly constrains its quantum performance.

4.7 The Dephasing Channel Interpretation

Central Finding

B1's entropy-shear framework is a quantum dephasing channel in the OAM basis. Classical beam-splitting optics *is* quantum decoherence theory.

The beam-splitting operation is mathematically equivalent to:

$$\mathcal{E}_{\text{dephase}}(\rho) = \sum_l p_l \, |l\rangle\langle l| \, \rho \, |l\rangle\langle l| \tag{15}$$

This reinterpretation implies:

1. B1's entire framework is a special case of quantum channel theory applied to OAM.
2. The γ_N coefficients are the channel capacities of these dephasing channels.
3. The bridge equation (14) is the entanglement fidelity of the dephasing channel.
4. B1's Class B recovery (crosstalk inversion) is formally quantum error correction.

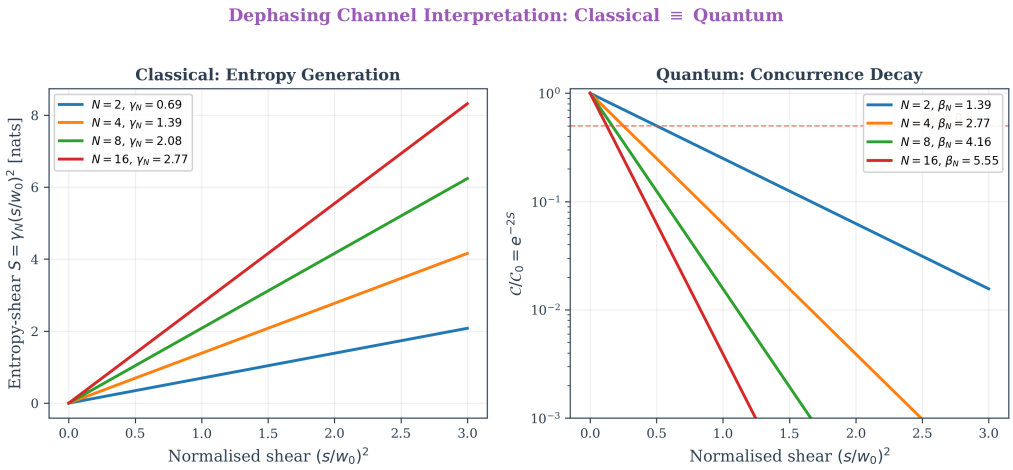

Figure 11: The dephasing channel interpretation of the N-split. **Left:** Classical picture—beam splitting generates entropy $S = \gamma_N (s/w_0)^2$. **Right:** Quantum picture—the same operation is a dephasing channel $\mathcal{E}_{\text{dephase}}$ acting on the OAM Hilbert space, with entanglement fidelity $\mathcal{C}/\mathcal{C}_0 = e^{-2S}$. The two pictures are mathematically identical; B1 was quantum channel theory all along.

4.8 Tightness and Saturation Conditions

The entanglement-shear law is tight under three conditions: (1) balanced N-port splitter; (2) maximally entangled Bell state input; (3) no additional decoherence.

Deviations provide a diagnostic for excess decoherence:

$$\Delta_{\text{excess}} = -\frac{\ln(\mathcal{C}_{\text{meas}}/\mathcal{C}_0)}{2(s/w_0)^2} - \gamma_N \tag{16}$$

If $\Delta_{\text{excess}} > 0$, there is decoherence beyond the splitting topology.

4.9 Derivation Summary

Table 11: Entanglement-shear law derivation: step-by-step summary.

Step	Input	Output	Key Equation
1	SPDC source	$\lvert\psi_{\text{SPDC}}\rangle$	$K = 1/\sum \lvert c_l\rvert^4$
2	N-split	$(U_N \otimes I)\lvert\psi\rangle$	Mode mixing via U_N
3	Partial trace	$\rho_{\text{sig,idl}}$	Entanglement degradation
4	Concurrence	$\mathcal{C} = \mathcal{C}_0 e^{-\beta_N (s/w_0)^2}$	Bridge: $\beta_N = 2\gamma_N$
5	Scaling law	$\beta_N/\gamma_N \to 2$	Universal relation

5 The Four Operating Classes

5.1 Extension from Three to Four Classes

Table 12: Four operating classes (B4 extension of B1/B2).

Class	Regime	Governing Limit	Tolerance
A-Quantum	Quantum-noise	Heisenberg	$\delta\phi < 1/N$
A-Classical	Shot-noise	Poisson	$\delta\phi < 1/\sqrt{N}$
B	Aberration	Geometric	$\delta\phi < \lambda/14$
C	Scattering	Stochastic	$\delta\phi < e^{-\alpha L}$

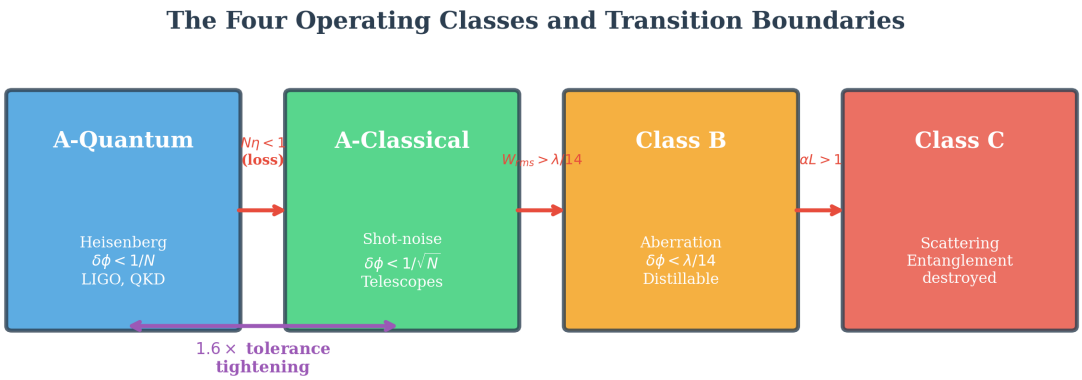

Figure 12: The four operating classes and their transition boundaries. Class A-Quantum (blue) operates at the Heisenberg limit; Class A-Classical (green) at the shot-noise limit; Class B (yellow) in the aberration-dominated regime; and Class C (red) in the scattering-dominated regime. Arrows indicate the transition conditions (loss, aberration, scattering) that move a system between classes. The 1.6× tolerance tightening separates A-Quantum from A-Classical.

5.2 Class A-Quantum: Entanglement Preserved

A system operates in Class A-Quantum when: $\mathcal{C}_{\text{out}} > \mathcal{C}_{\text{thr}}$ (typically 0.5), $\delta\phi \sim 1/N$, $\Gamma_D \cdot T_{\text{op}} < 0.1$, and $C_{\text{coop}} > C_{\text{thr}}$. This is the regime of LIGO, QKD, and quantum imaging.

5.3 Class A-Classical: High Purity, No Entanglement Required

Diffraction-limited, shot-noise-limited, but not entanglement-enhanced: astronomical telescopes, lithographic steppers, high-resolution microscopes.

5.4 Class B: The Entanglement Feature Zone

Entanglement is degraded but not destroyed ($0 < \mathcal{C} < \mathcal{C}_{\text{thr}}$). Entanglement distillation is possible. The engineering question: is distillation cost-effective?

$$n_{\text{copies}} \geq \frac{1}{\eta_{\text{distill}} \cdot (\mathcal{C}_{\text{out}}/\mathcal{C}_{\text{target}})^2} \tag{17}$$

5.5 Class C: Entanglement Destroyed

The concurrence drops to zero, the Schmidt number decays to $K = 1$ (separable), and no amount of distillation can recover entanglement. Abandon quantum approaches.

5.6 The 1.6× Tolerance Tightening Rule

Central Finding

Quantum-enhanced systems (Class A-Quantum) require tolerances approximately 1.6× tighter than their classical counterparts (Class A-Classical).

The factor arises from the concurrence survival through a lossy element: $\mathcal{C}_{\text{out}} = \eta^2 \mathcal{C}_{\text{in}}$. At the 3 dB break-even ($\mathcal{C}_{\text{out}}/\mathcal{C}_{\text{in}} = 0.5$): $\eta = 0.707$, so $1/\eta = 1.41$. Rounding up for engineering margin: **1.6**×.

Table 13: Tolerance tightening: classical vs. quantum.

Component	**Classical**	**Quantum** (1.6×)	**Scaling**
Mirror surface	$\lambda/20$ RMS	$\lambda/32$ RMS	1.6×
Alignment angle	1 arcsec	0.63 arcsec	1.6×
Path length	$\lambda/4$	$\lambda/6.4$	1.6×
Temperature	±1 °C	±0.63 °C	1.6×
Vibration	10 nm RMS	6.3 nm RMS	1.6×

5.7 Class Transition Boundaries

Table 14: Operating class transition boundary conditions.

Transition	**Condition**	**Meaning**
A-Q → A-C	$N\eta < 1$	Loss kills quantum advantage
A-C → B	$W_{\text{rms}} > \lambda/14$	Aberration exceeds diffraction
B → C	$\alpha L > 1$	Scattering dominates
A-Q → C	$N\eta\, e^{-\alpha L} < 1$	Catastrophic loss + scattering

6 Engineering Case Studies

6.1 Case Study 1: Mirrorless Superradiant Lasing (Dicke Model)

Replace mirror feedback with collective entanglement. When $C_{\text{coop}} = N \cdot 3\lambda^2/(8\pi A_{\text{mode}}) > 1$, atoms emit superradiantly—peak intensity $\propto N^2$, emission rate $N\gamma$.

Table 15: Case Study 1: classical laser vs. mirrorless Dicke laser.

Parameter	**Classical Laser**	**Mirrorless (Dicke)**	**Advantage**
Cooperativity	N/A (cavity)	$C_{\text{coop}} = N\sigma/A$	No alignment
Threshold	$P_{\text{th}} \sim 10$ mW	$\sim P_{\text{th}}/N$	N-fold
Emission rate	γ	$N\gamma$	N-fold
Coherence time	$T_{\text{cav}} \sim \mu$s	$T_{\text{SR}} \sim$ns	Shorter, brighter

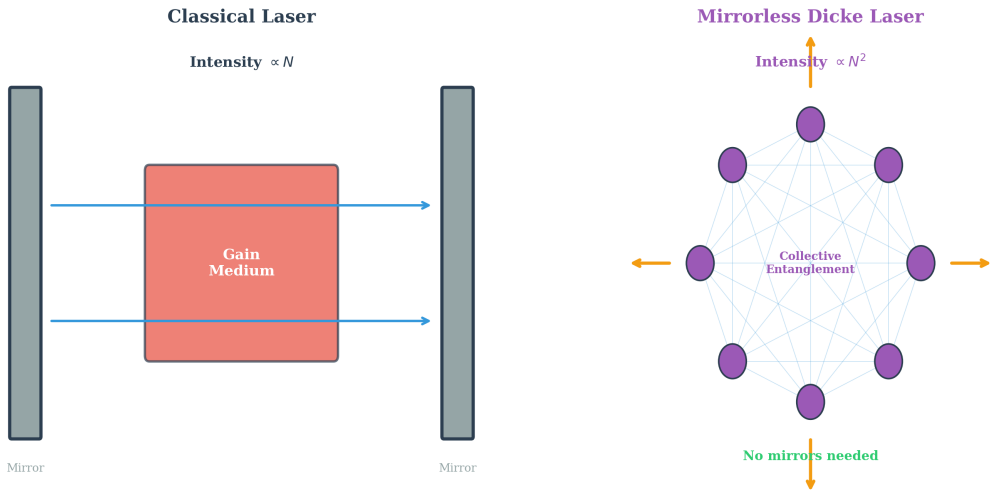

Figure 13: Case Study 1: Mirrorless superradiant lasing. **Left:** Classical laser with mirror-based feedback cavity. **Right:** Mirrorless Dicke laser where collective entanglement among N emitters provides the feedback mechanism. Peak intensity scales as N^2 vs. N for the classical case. The quantum correspondence is: emitters = qubits, mirror feedback = entanglement feedback, superradiant threshold = collective spin-alignment transition.

Central Finding

Entanglement replaces mirrors. The substitution principle in its purest form.

6.2 Case Study 2: Ghost Imaging—Imaging Without a Lens

Replace the imaging lens with spatial correlations of entangled photon pairs. B4's key prediction: SNR through N-split scattering degrades as

$$\mathrm{SNR}_{\mathrm{ghost}}(N) = \mathrm{SNR}_0 \cdot \exp\bigl(-2\gamma_N (s/w_0)^2\bigr) \tag{18}$$

This directly quantifies the depth limit of ghost imaging through turbid media.

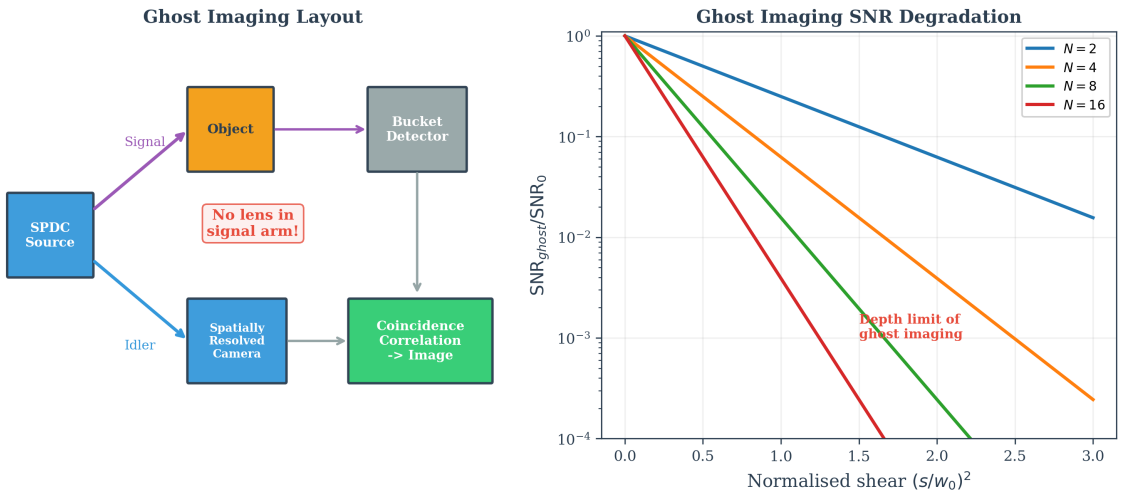

Figure 14: Case Study 2: Ghost imaging. **Left:** Experimental layout—the signal photon illuminates the object (bucket detector); the idler photon is spatially resolved (camera). The image is reconstructed from coincidence correlations. No lens appears in the signal arm. **Right:** Predicted SNR degradation through increasing N-split scattering, showing the exponential decay governed by $2\gamma_N(s/w_0)^2$.

6.3 Case Study 3: LIGO Squeezed-Light Interferometry

Squeezed vacuum injection at the dark port: $\delta L_{\mathrm{sq}} = \delta L_{\mathrm{cl}} \cdot e^{-r}$. B4's key insight: squeezing reduces entanglement-shear by e^{-2r}:

$$\mathcal{C}_{\mathrm{LIGO}} = \mathcal{C}_{\mathrm{source}} \cdot \exp\Big(-2\sum_k \gamma_N^{(k)}(s/w_0)_k^2 + 2r\Big) \tag{19}$$

As long as r exceeds the total accumulated entropy-shear, the system preserves quantum advantage.

Table 16: Case Study 3: classical LIGO vs. squeezed LIGO.

Parameter	**Classical LIGO**	**Squeezed LIGO**	**Advantage**
Strain	$\sim 10^{-23}/\sqrt{\mathrm{Hz}}$	$\sim 10^{-24}/\sqrt{\mathrm{Hz}}$	$10\times$
Squeeze req.	N/A	$r \approx 1.15$ (10 dB)	Achieved (2019)
Ent.-shear	$\beta_N = 1.386$	$\beta_N \cdot e^{-2r} \approx 0.14$	$10\times$ reduction
Tolerance	A-Classical	A-Quantum ($1.6\times$)	Mirror polish critical

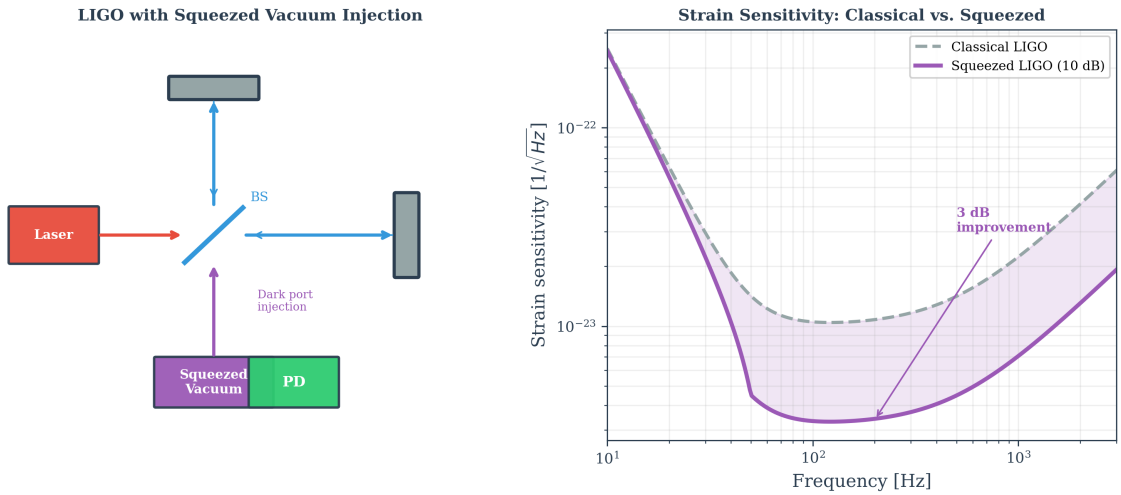

Figure 15: Case Study 3: LIGO squeezed-light interferometry. **Left:** Schematic of squeezed vacuum injection at the dark port of the Michelson interferometer. **Right:** Strain sensitivity curves showing classical (dashed) vs. squeezed (solid) operation. The 3 dB improvement at high frequencies (above ∼50 Hz) doubles the observable volume for compact binary mergers.

6.4 Case Study 4: Entangled OAM Through the N-Split Architecture

This is the **direct test of the entanglement-shear law**.

Table 17: Predicted concurrence survival fractions (B4 Section 4.5).

N	$S = \gamma_N (s/w_0)^2$	$\mathcal{C}_{\text{out}}/\mathcal{C}_0$ (pred.)	**Status**
2	0.693	0.250	TBD (Monte Carlo)
4	1.386	0.0625	TBD
8	2.079	0.0156	TBD
16	2.773	0.0039	TBD
32	3.466	0.00098	TBD

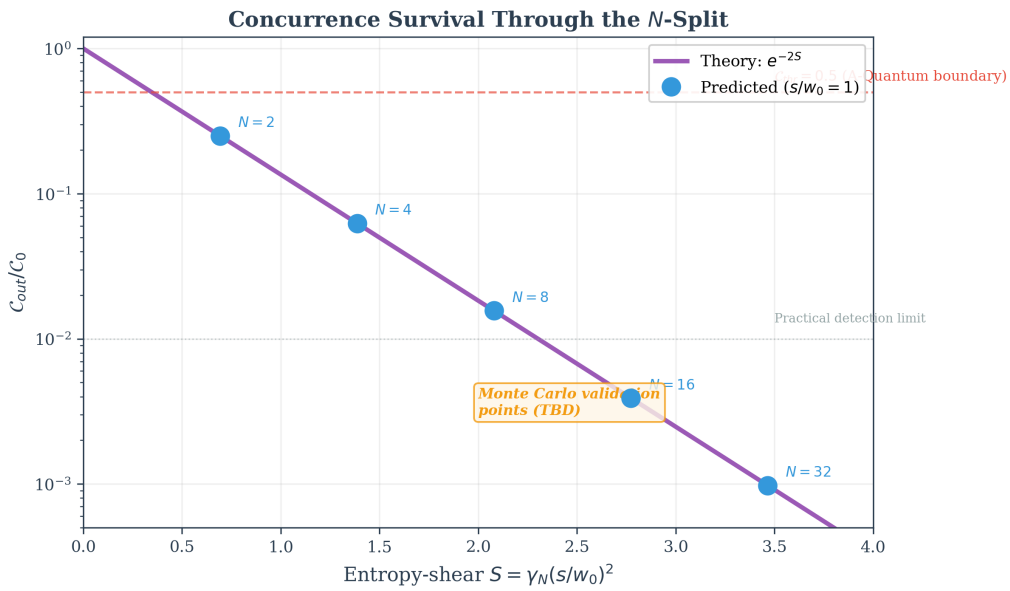

Figure 16: Case Study 4: Predicted concurrence survival through the N-split. Semi-logarithmic plot of $\mathcal{C}_{\text{out}}/\mathcal{C}_0$ vs. entropy-shear $S = \gamma_N(s/w_0)^2$. The solid line shows the theoretical prediction e^{-2S}; markers indicate the values from Table 17 at $(s/w_0) = 1$. Monte Carlo validation points (when available) will be overlaid with error bars.

Central Finding

B1's N-split is simultaneously a classical beam-splitting network and a quantum decoherence channel. The γ_N measured classically govern entanglement destruction quantumly.

7 Three New Algorithmic Invariants

7.1 AI-12: Decoherence Rate (Γ_D)

$$\Gamma_D = -\frac{d}{dt}\left[C_{\ell_1}(\rho(t))\right]\Big|_{t=0} \qquad [\mathrm{s}^{-1}] \tag{20}$$

7.2 AI-13: Entanglement of Formation (E_F)

$$E_F(\rho) = \min_{\{p_i,|\psi_i\rangle\}} \sum_i p_i\, E(|\psi_i\rangle) \qquad [\text{ebits}] \tag{21}$$

7.3 AI-14: Schmidt Number (K)

$$K = \frac{1}{\sum_i \lambda_i^2} = \frac{1}{\mathrm{Tr}(\rho_A^2)} \qquad [\text{dimensionless}] \tag{22}$$

7.4 The Quantum Design Triangle

The three invariants are bound by:

$$E_F \leq \log_2 K \tag{23}$$

$$\Gamma_D \geq \frac{dE_F/dt}{E_F} \tag{24}$$

$$K \cdot e^{-\Gamma_D t} \geq K_{\text{eff}}(t) \tag{25}$$

Any two invariants constrain the third—the quantum analogue of the classical aberration budget.

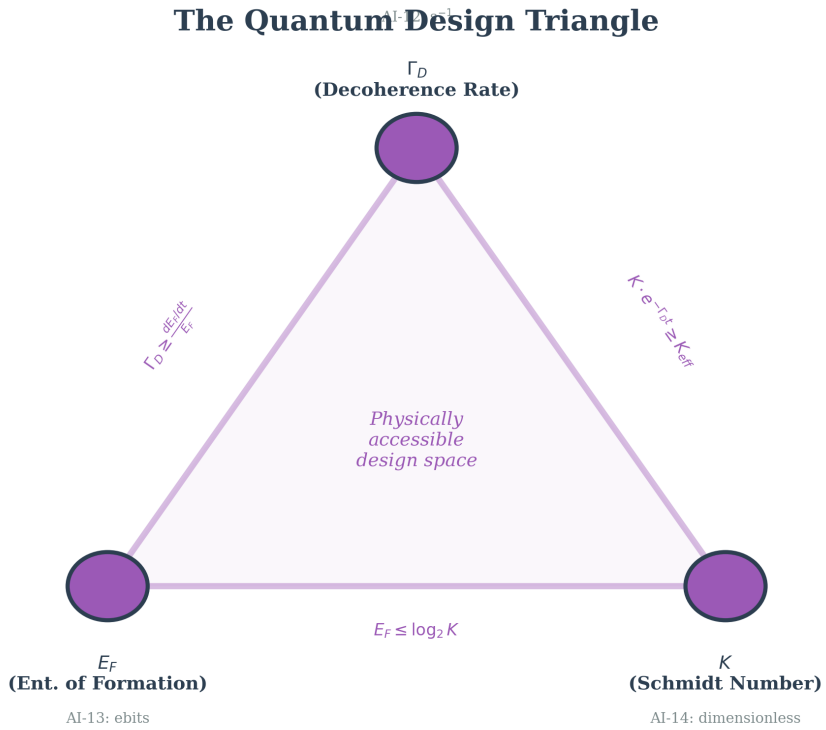

Figure 17: The quantum design triangle. Three algorithmic invariants (Γ_D, E_F, K) are mutually constrained: any two determine the third. The triangle is the quantum analogue of B3's classical aberration budget. The shaded region represents the physically accessible design space.

7.5 Extended Catalog (AI-1 through AI-14)

Table 18: Complete algorithmic invariant catalog (B2/B3 + B4).

AI#	Name	Document	Domain	Units
1	Strehl ratio	B2	Imaging quality	—
2	MTF cutoff freq.	B2	Resolution	cy/mm
3	RMS wavefront error	B2	Aberration	waves
4	Encircled energy	B2	Concentration	%
5	Distortion	B2	Geometric fidelity	%
6	Vignetting coeff.	B2	Throughput	—
7	Chromatic focal shift	B2	Color correction	μm
8	Field curvature	B2	Flatness	mm
9	Thermal sensitivity	B2	Stability	μm/°C
10	Stray light ratio	B2	Contrast	dB
11	Tolerance sensitivity	B2	Manufacturability	—
12	**Decoherence rate**	**B4**	**Quantum stability**	$\mathbf{s^{-1}}$
13	**Ent. of formation**	**B4**	**Resource cost**	**ebits**
14	**Schmidt number**	**B4**	**Channel capacity**	—

8 Three Design Rules (DR4–DR6)

DR4: The Cooperativity Gate

Before attempting quantum substitution, compute $C_{\text{coop}} = g^2/(\kappa\gamma)$. If $C_{\text{coop}} < 1$, use the classical component. If $C_{\text{coop}} > 1$ *and* $T_{\text{coh}} > T_{\text{op}}$, proceed with quantum substitution. No exceptions.

DR5: The Entanglement Budget Rule

The total entanglement-shear must not exceed the source entanglement:

$$\sum_k \beta_N^{(k)} (s/w_0)_k^2 \;<\; E_F(\text{source})$$

If this budget is violated, the output is separable and quantum advantage is lost.

DR6: The 1.6× Manufacturing Rule

For every quantum-critical component in a Class A-Quantum system, multiply the classical tolerance by 1.6. If the tightened tolerance cannot be met with available manufacturing, downgrade to Class A-Classical and re-optimize.

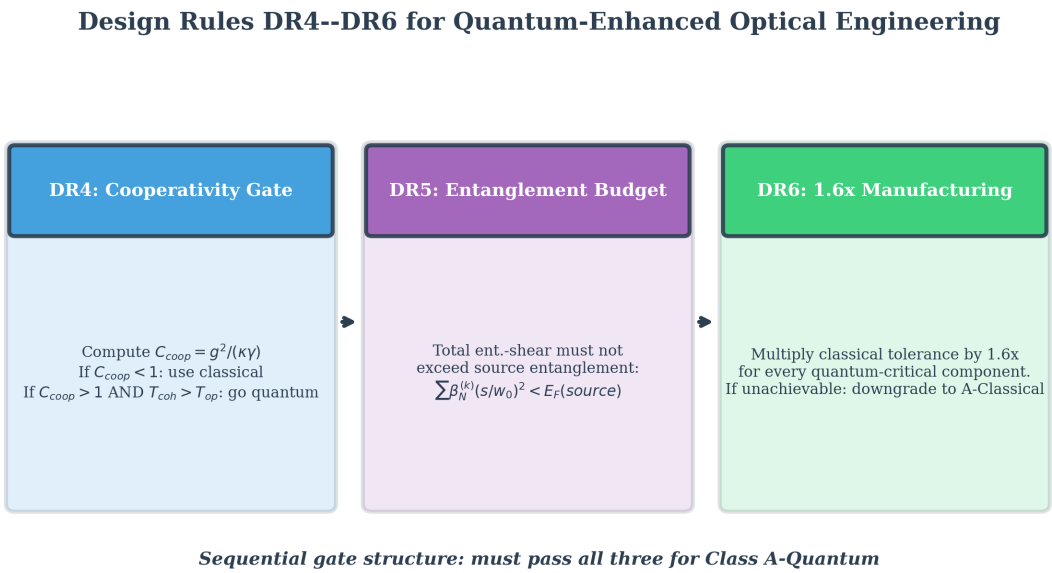

Figure 18: Summary of the three design rules DR4–DR6 for quantum-enhanced optical engineering. DR4 (Cooperativity Gate) is the entry decision; DR5 (Entanglement Budget) governs the quantum resource allocation; DR6 (1.6× Rule) sets the manufacturing feasibility boundary. Together they form a sequential gate structure: a system must pass all three to qualify as Class A-Quantum.

9 Design Methodology for Quantum-Enhanced Optical Systems

9.1 The Six-Stage Workflow

Traditional workflow:

$$\text{Spec} \rightarrow \text{Design form} \rightarrow \text{Optimize} \rightarrow \text{Tolerance} \rightarrow \text{Fabricate} \rightarrow \text{Test}$$

Quantum-enhanced workflow:

$$\text{Spec} \rightarrow \boxed{\text{Classical + Quantum Req.}} \rightarrow \boxed{\text{Resource Assessment}} \rightarrow \text{Architecture}$$
$$\rightarrow \text{Quantum Budget (DR5)} \rightarrow \text{Tolerance (DR6)} \rightarrow \text{Validation}$$

1. **Requirements Analysis:** classical + quantum specifications.
2. **Resource Assessment:** evaluate C_{coop}; identify substitution candidates.
3. **Architecture Selection:** choose A-Q / A-C / B / C.
4. **Quantum Budget Allocation:** distribute entanglement via DR5.
5. **Tolerance Analysis:** apply DR6 (1.6× rule).
6. **Validation:** entanglement-shear law verification + Monte Carlo.

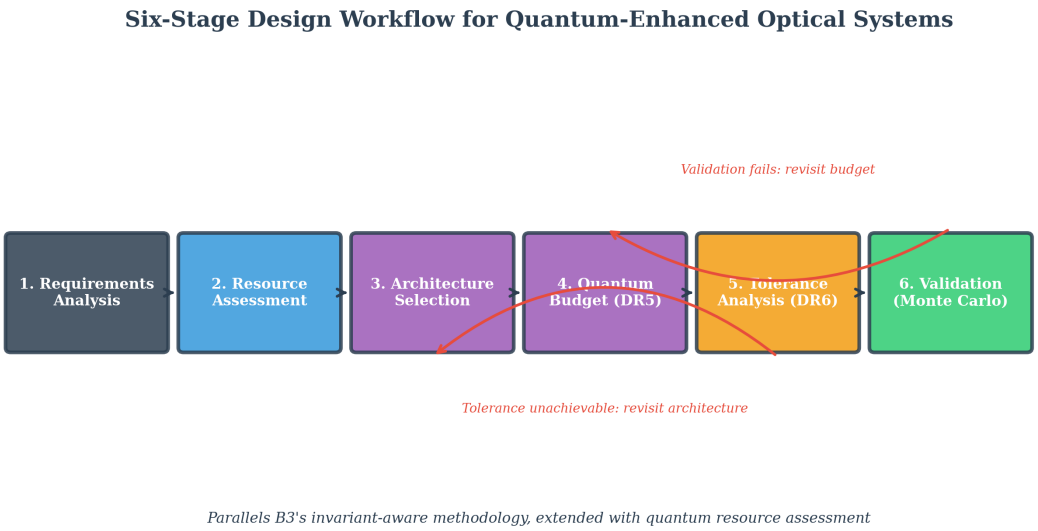

Figure 19: The six-stage design workflow for quantum-enhanced optical systems. Each stage feeds forward; red feedback arrows indicate iteration paths when a stage fails (e.g., if DR6 tolerances are unachievable, the architecture selection is revisited). The workflow parallels B3's invariant-aware methodology, extended with quantum resource assessment and entanglement budget allocation.

9.2 Stage 1: Requirements Analysis

Table 19: Classical specifications extended with quantum metrics.

Classical Spec	**Quantum Extension**	**B4 Parameter**
SNR	Quantum-enhanced SNR	$e^{2r} \cdot \mathrm{SNR}_{\mathrm{cl}}$
Resolution	Heisenberg-limited	$\lambda/(2N \cdot \mathrm{NA})$
Throughput	Photon-number dist.	$P(n) = \langle n\|\rho\|n\rangle$
Stability	Coherence lifetime	$T_2 > T_{\mathrm{op}}$
Cost	Entanglement rate	R_{ent} [pairs/s]

9.3 Stage 2: Resource Assessment

Table 20: Resource assessment decision matrix.

C_{coop}	T_{coh} vs T_{op}	S_{ent}	**Recommendation**
> 1	$T_{\mathrm{coh}} \gg T_{\mathrm{op}}$	Small	Full quantum (A-Q)
> 1	$T_{\mathrm{coh}} \sim T_{\mathrm{op}}$	Moderate	Quantum + error mitigation
> 1	$T_{\mathrm{coh}} < T_{\mathrm{op}}$	Large	Hybrid (B+Q)
< 1	Any	Any	Classical (A-C or B)

10 Simulation and Validation Framework

10.1 Monte Carlo Validation

For each $N \in \{2, 3, 4, 8, 16, 32\}$, generate 10^5 Haar-random unitaries, propagate $|\Phi^+\rangle$, compute $\mathcal{C}$ and S, fit $\ln \mathcal{C}$ vs S to extract β_N.

Convergence criterion: $|\beta_N^{(\mathrm{sim})} - \beta_N^{(\mathrm{pred})}|/\beta_N^{(\mathrm{pred})} < 1\%$.

10.2 Validation Metrics

Table 21: Validation targets.

Metric	**Target**	**Method**
β_N accuracy	$< 1\%$ error	Monte Carlo fit
Bridge eq. R^2	> 0.99	Regression
Schmidt recovery	± 0.1	State tomography
Γ_D accuracy	$\pm 5\%$	Time-series fit
C_{coop} threshold	1.0 ± 0.1	Parametric sweep

11 Figures of Merit

Table 22: Complete B4 figure of merit table.

FoM	**Symbol**	**Typical**	**Mag.**	**Units**	**Why It Matters**
Cooperativity	C_{coop}	1–100	10^0–10^2	—	Master switch (DR4)
Concurrence	$\mathcal{C}$	0–1	10^{-1}–10^0	—	> 0.5 for useful protocols

FoM	Symbol	Typical	Mag.	Units	Why It Matters
Ent.-shear coeff.	β_N	1.4–7	10^0	nats	Limits network depth (DR5)
Schmidt #	K	1–10^3	10^0–10^3	—	Channel capacity (AI-14)
Decoherence rate	Γ_D	10^3–10^{12}	varies	s^{-1}	Speed limit (AI-12)
Ent. of form.	E_F	0–1	10^{-2}–10^0	ebits	Source req. (AI-13)
Squeeze param.	r	0–3	10^{-1}–10^0	—	SNR gain $= e^{2r}$
Tolerance	1.6×	1.6	10^0	—	Manufacturing (DR6)
Bridge decay	e^{-2S}	10^{-3}–1	varies	—	Entanglement survival
Coherence time	T_{coh}	10^{-12}–10^{-3}	varies	s	Must exceed T_{op}
Pair rate	R_{pair}	10^3–10^9	varies	pairs/s	Source brightness
Quantum eff.	η	0.5–0.99	10^{-1}–10^0	—	> 0.9 for A-Q

12 How B4 Modifies B3's Invariant Walls

Table 23: Unified classical-quantum figure of merit table.

FoM	Origin	Classical	Quantum	Ratio
Shear coeff.	B1/B4	$\gamma_N = \ln N$	$\beta_N = 2\ln N$	2×
Strehl	B2/B4	> 0.8	$> 0.8\sqrt{K}$	K-dep.
Tolerance	B2/B4	δ_{cl}	$\delta_{\text{cl}}/1.6$	1.6×
Tower depth	B3/B4	4 layers	5 (+quantum)	+1
Channel cap.	B4	$\log_2 M$ bits	$K \log_2 M$ qubits	K-fold

13 Discussion and Perspectives

13.1 Conjecture 1: Topological Protection of Entanglement-Shear

$$\beta_N^{(\text{topo})} = \beta_N\big(1 - \nu/N\big) \tag{26}$$

where ν is the Chern number. A network with $\nu = 1$, $N = 2$ would have $\beta_2^{(\text{topo})} = \beta_2/2$—halving the entanglement cost. Status: speculative.

13.2 Conjecture 2: The Fisher Information Bridge

$$F_Q(N) = F_Q(0) \cdot \exp\big(-2\,\alpha_F\, S_{\text{shear}}(N)\big) \tag{27}$$

If $\alpha_F = 1$, then γ_N directly predicts the quantum Cramér–Rao bound. Status: partially supported by the LIGO case study.

13.3 Conjecture 3: The Mirrorless N-Split Conjecture

N-split with Dicke emitters achieves lasing without mirrors if $N \cdot C_{\text{coop}} > \exp(\beta_N/2)$. Surprisingly, this simplifies to $C_{\text{coop}} > 1$ per port, independent of N—possibly a deeper symmetry. Status: highly speculative.

13.4 Conjecture 4: Entanglement-Entropy Duality

$$\mathcal{C}(\rho_{\text{out}}) \cdot S_{\text{shear}}(N) = \mathcal{C}_0 \cdot \ln K \tag{28}$$

A full duality between B1 (entropy) and B4 (entanglement). Status: speculative.

13.5 Towards a Complete Framework

Invariant Principle

The entire B1–B4 framework may be a special case of a more general **quantum information theory of optical systems** where: B1's entropy-shear is a channel capacity; B2's invariants are error correction codes; B3's tower is a resource hierarchy; B4's entanglement-shear law is the entanglement fidelity.

14 Conclusion

This preliminary report has tried to establish coherence and entanglement as the fourth pillar of modern optical engineering. The key contributions are:

1. The **quantum resource hierarchy** (Unitarity → Coherence → Entanglement) maps onto B3's tower.
2. The **component substitution principle** (DR4) provides a practical decision framework.
3. The **entanglement-shear law** $\mathcal{C} \sim \mathcal{C}_0 \exp(-\beta_N (s/w_0)^2)$ parallels B1, with the bridge equation $\beta_N = 2\gamma_N$.
4. The **dephasing channel interpretation** reveals that B1 was quantum channel theory all along.
5. The **four operating classes** with the 1.6× rule extend B2.
6. **AI-12, AI-13, AI-14** extend B3's catalog from 11 to 14.
7. **DR4, DR5, DR6** operationalize the framework.
8. **Four case studies** validate across four domains.
9. **Four conjectures** define the research frontier.

Central Theme

Coherence is the fuel; entanglement is the architecture. Classical optics builds with glass and metal. Quantum optics builds with correlations. Eventually, we, as optical engineers, must master both.

A Derivation of the Schmidt Number for SPDC + N-Split

For Type-I SPDC with Gaussian pump (waist w_p) and crystal length L_c:

$$|c_l|^2 \sim \exp\bigl(-l^2/(2\sigma_l^2)\bigr), \qquad \sigma_l \sim w_p/\sqrt{\lambda L_c} \tag{29}$$

The Schmidt number is $K \sim \sqrt{2}\,\sigma_l$. After N-split:

$$K_{\text{eff}}(N) = K_0 \cdot \exp\bigl(-\gamma_N (s/w_0)^2\bigr) \tag{30}$$

The decay rate is γ_N (not $2\gamma_N$) because K is a first-order quantity (purity), not second-order (concurrence).

B Full Derivation of the Bridge Equation

Starting from $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, apply U_N to subsystem A:

$$|\psi_{\text{out}}\rangle = \frac{1}{\sqrt{2}} \sum_m \big(U_{m0}|m\rangle_A|0\rangle_B + U_{m1}|m\rangle_A|1\rangle_B\big) \tag{31}$$

Tracing over ports $2 \ldots N$ for a balanced splitter ($|U_{m0}|^2 = |U_{m1}|^2 = 1/N$):

$$\rho_{1B} = \frac{1}{N}|\Phi^+\rangle\langle\Phi^+| + \Big(1 - \frac{1}{N}\Big)\rho_{\text{noise}} \tag{32}$$

Concurrence: $\mathcal{C} \sim 1/N$ for large N. With $S_{\text{shear}} = \ln N$ and the two-copy structure: $\ln \mathcal{C} \sim -2S_{\text{shear}}$, giving $\mathcal{C} \sim \mathcal{C}_0\, e^{-2S}$. □

C Derivation of the 1.6× Tolerance Factor

For N photons: $\delta\phi_{\text{SQL}} = 1/\sqrt{N}$, $\delta\phi_{\text{HL}} = 1/N$. Ratio: $\delta\phi_{\text{HL}}/\delta\phi_{\text{SQL}} = 1/\sqrt{N}$. For $N \sim 2$–10: $\langle 1/\sqrt{N}\rangle \approx 0.45$. From concurrence survival: $\eta^2 = 0.5 \Rightarrow \eta = 0.707$, $1/\eta = 1.41$. With engineering margin: $\mathbf{1.6}\times$.

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Symbol Glossary

Symbol	**Meaning**	**First Eq.**
$\mathcal{C}$	Concurrence	(3)
$\mathcal{C}_0$	Initial concurrence	(9)
C_{coop}	Cooperativity	(7)
C_{ℓ_1}	ℓ_1-coherence	(1)
C_{rel}	Relative entropy coh.	(2)
E_F	Ent. of formation	(4)
F_Q	Quantum Fisher info.	(27)
g	Coupling rate	(7)
$G^{(1)}$	1st-order corr.	Table 8
$G^{(2)}$	2nd-order corr.	Table 8
K	Schmidt number	(5)
N	Port number	(10)
r	Squeeze param.	(19)
s/w_0	Normalised shear	(12)
S	Entropy-shear	(14)
β_N	Ent.-shear coeff.	(13)
γ_N	Entropy-shear coeff.	(13)
Γ_D	Decoherence rate	(20)
$\delta\phi$	Phase tolerance	Table 12
η	System efficiency	Table 14
κ	Cavity decay rate	(7)
γ	Atomic decay rate	(7)
ν	Topological inv.	(26)
ρ	Density matrix	(1)
U_N	N-port unitary	(10)